先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 09 – Support Vector Machines for Classification and Regression



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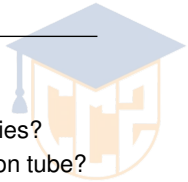
Advanced Computing and Digital Technology Research Center

University of Chinese Academy of Sciences

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SVM syllabus blocks and timing

Syllabus block	Time	Driving question
Linear SVM	0.5 h	Which separating line is safest?
Soft margin	0.5 h	How much violation should be tolerated?
Kernel methods	1.0 h	How can similarity create nonlinear boundaries?
Support vector regression	1.0 h	Which residuals should constrain a regression tube?



A margin story from inspection

Factory gate

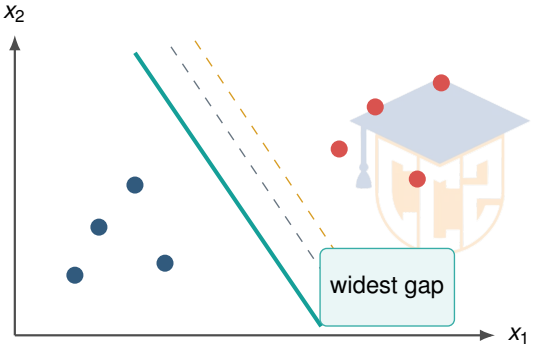
Two acceptable product families can be separated by several straight lines.

Tomorrow

A measurement may shift slightly after sensor recalibration.

Choice

A boundary with no clearance can fail after a tiny shift.



The lecture follows one margin principle



One margin principle; several computational views.

1. Choose a geometric separation.
2. Permit measured violations when data are noisy.
3. Rewrite the solution with sample similarities.
4. Replace inner products by kernels.
5. Use an epsilon tube for regression.

A hyperplane separates two classes

$$f(x) = w^T x + b, \qquad \widehat{y} = \text{sign}(f(x))$$

Read it

A two-feature defect dataset becomes a line on the screen.

Boundary

The score is not yet a calibrated probability.

The weight vector is normal to the boundary

Core idea

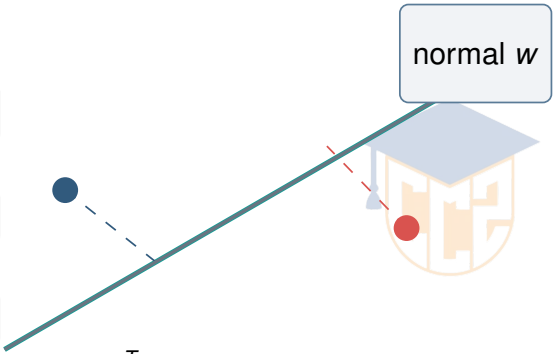
w points perpendicular to $w^T x + b = 0$.

Concrete case

Project a sample onto the normal to measure its signed position.

Watch for

Reading coefficient size without feature scaling distorts the geometry.



$w^T x + b$: signed projection

Distance to a hyperplane needs normalization

$$\text{dist}(x, H) = \frac{|w^T x + b|}{\|w\|_2}$$

Read it

Doubling both w and b leaves the boundary unchanged.

Boundary

The raw score alone has arbitrary scale.

Labels make one signed constraint

$$y_i(w^T x_i + b) > 0, \qquad y_i \in \{-1, +1\}$$

Read it

Both classes satisfy the same inequality after multiplication by the label.

Boundary

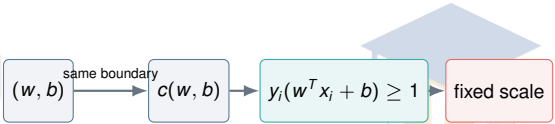
A zero score lies exactly on the decision boundary.

Parameter rescaling fixes the functional margin

$$y_i(w^T x_i + b) \geq 1$$

Read it

Scale a separating solution until the closest case reaches one.



Boundary

The number one is a normalization convention, not a physical distance.

Two supporting planes define the corridor

$$w^T x + b = +1 \quad \text{and} \quad w^T x + b = -1$$

Read it

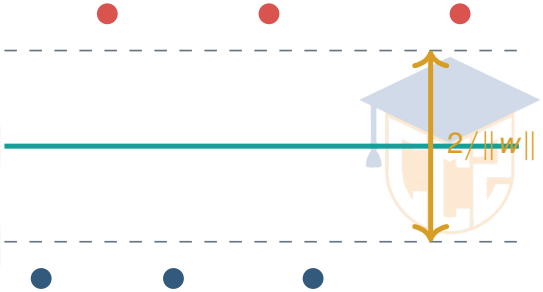
The decision plane lies halfway between the two supports.

Boundary

Moving only one support plane would break the common parameterization.

The full margin width is two over the norm

$$\rho = \frac{2}{\|w\|_2}$$



Read it

A smaller weight norm produces a wider corridor after normalization.

Boundary

The formula depends on the fixed functional-margin convention.

Hard margin is a constrained optimization

$$\min_{w,b} \frac{1}{2}\|w\|_2^2 \quad \text{s.t. } y_i(w^T x_i + b) \geq 1$$

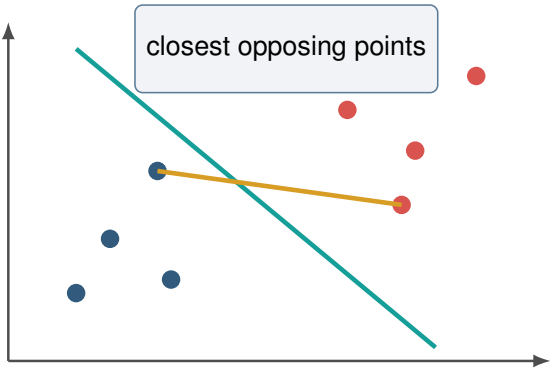
Read it

Every training case must be correctly classified outside the margin.

Boundary

One contradictory or mislabeled case makes the problem infeasible.

Only the closest cases set the solution



Start from evidence

Move a point that touches a supporting plane and the boundary may rotate.

Model statement

Moving a distant point slightly does not change the optimum.

Failure mode

A visually extreme point is not necessarily a support vector.

Support vectors encode local boundary evidence

What changes

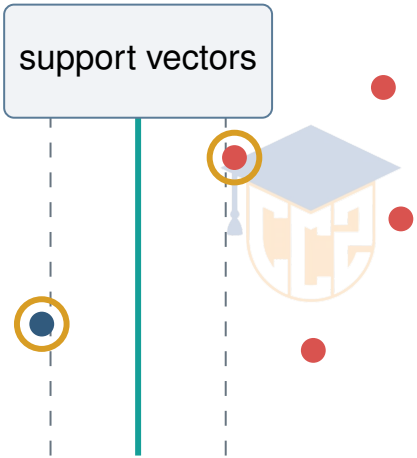
The classifier is determined by a subset near the class frontier.

What can break

Small support count does not automatically imply good generalization.

Try this

A large dataset can yield a small support set.



Feature scaling changes the margin problem

Core idea

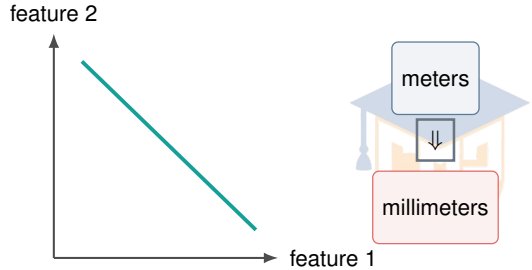
Standardize continuous features before distance-based margin fitting.

Concrete case

One feature measured in millimeters can dominate another measured in meters.

Watch for

Fit the scaler only on the training fold.



Rescaling a feature changes Euclidean geometry and therefore the margin.

A vertical separator can be solved by inspection

$$w = (1, 0)^T,\ b = -3 \Rightarrow x_1 = 3$$

Read it

Blue cases lie left of three and red cases lie right of three.

Boundary

Many scaled parameter pairs represent the same line.

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Compute a margin from two nearest coordinates

$$x_1 = 2 \text{ and } x_1 = 4 \Rightarrow \rho = 2$$

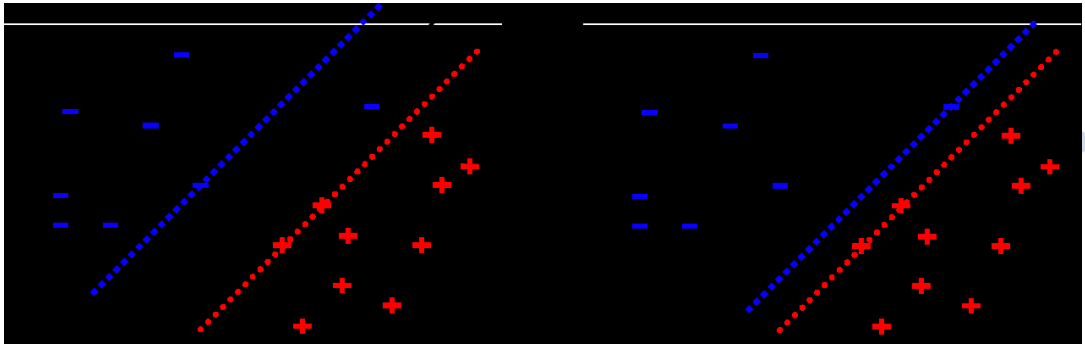
Read it

The midpoint boundary is $x_1 = 3$.

Boundary

This shortcut works only when the closest supporting planes are parallel coordinate lines.

Compare separating margins



A larger margin may require tolerating some training errors.

Large margin is a stability heuristic

What changes

Small perturbations must travel farther before crossing the boundary.

What can break

A large geometric margin cannot protect against arbitrary distribution shift.

Try this

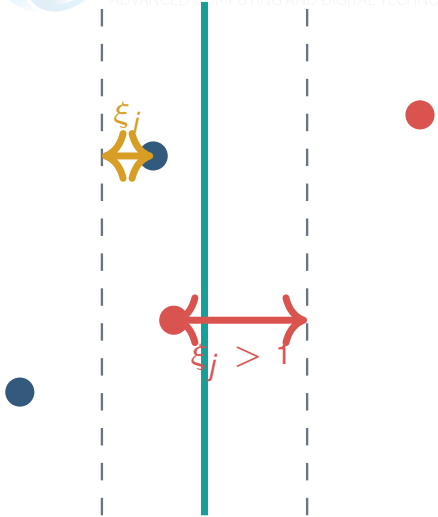
A recalibrated sensor shifts every measurement by a tiny amount.

Hard margin assumes exact separability

Lens	Evidence
Principle	There exists at least one hyperplane satisfying every unit-margin constraint.
Case	Clean synthetic points may meet this condition.
Risk	Real labels, overlapping classes, and measurement errors often violate it.



Slack variables measure margin violations



$$y_i(w^T x_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0$$

Read it

A correctly classified point inside the corridor has $0 < \xi_i < 1$.

Boundary

Slack is not simply the count of mistakes.

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Three slack regimes have different meanings

Slack	Position	Status
$\xi_i = 0$	On or outside support plane	No violation
$0 < \xi_i < 1$	Inside margin	Correct side
$\xi_i > 1$	Across boundary	Misclassified

Soft margin trades width for violations

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_i \xi_i$$

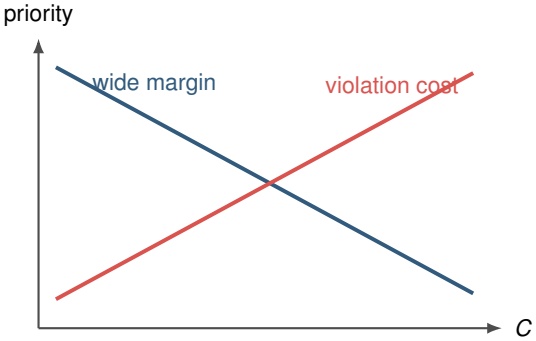
Read it

A noisy inspection label may be tolerated to keep a stable corridor.

Boundary

Large C is not always better.

The parameter C sets the violation price



Core idea

Small C favors a wider margin; large C emphasizes training violations.

Concrete case

Trace a validation curve over logarithmic values of C.

Watch for

The best value depends on scaling and sample size.

One outlier can bend a hard boundary

Core idea

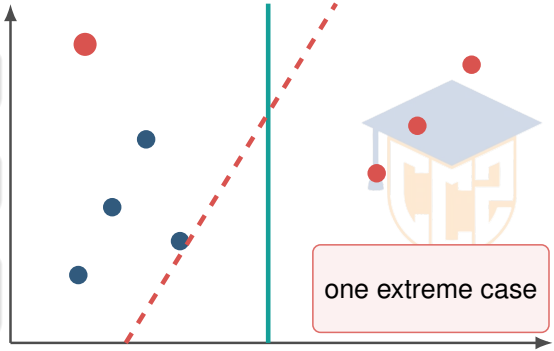
A very large C can give one contradictory point excessive influence.

Concrete case

Compare boundaries before and after adding one mislabeled case.

Watch for

Removing inconvenient points without an audit hides data quality problems.



Hinge loss removes explicit slack variables

$$\max(0, 1 - y_i f(x_i)) = \xi_i^*$$

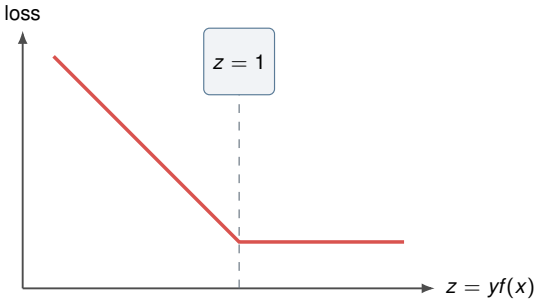
Read it

The optimal slack equals the positive part of the margin shortfall.

Boundary

Correct predictions inside the margin still incur loss.

Hinge loss is flat beyond the margin



Correct but inside the margin still pays loss.

$$\ell(z) = \max(0, 1 - z), \qquad z = yf(x)$$

Read it

A safely classified point with $z > 1$ adds no loss.

Boundary

A zero hinge loss is stronger than merely being correct.

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Compare zero-one, hinge, and logistic losses

What changes

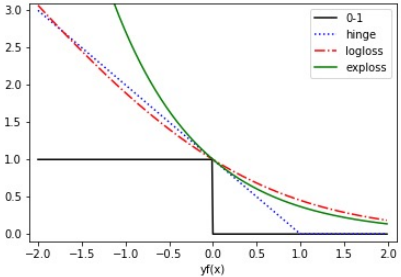
Hinge is a convex upper bound on zero-one loss.

What can break

Different surrogate losses imply different scores and calibration properties.

Try this

Logistic loss changes smoothly for every finite margin.



Class weights alter the cost of violations

$$\frac{1}{2}\|w\|^2 + \sum_i C_{y_i}\xi_i$$

Read it

Give rare defect cases a larger violation price.

Boundary

Weights should reflect costs and validation evidence, not only class counts.

Duplicate samples also change effective weights

Core idea

Repeating one observation multiplies its contribution to hinge loss.

Concrete case

A duplicated customer record can pull the boundary.

Watch for

Deduplication is part of model validity, not clerical cleanup.

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Regularization conventions can invert parameter names

$$\lambda\|w\|^2 + \frac{1}{N}\sum_i \ell_i \quad \text{versus} \quad \frac{1}{2}\|w\|^2 + C\sum_i \ell_i$$

Read it

Libraries may expose C while papers discuss λ.

Boundary

Do not compare raw numbers across conventions.

Validation chooses C, not training accuracy

Start from evidence

Select on the deployment-relevant metric.

Model statement

Use stratified or grouped cross-validation appropriate to the data source.

Failure mode

The final test set must remain outside the search.

A log scale reveals the useful C range

$$C \in \{10^{-4}, 10^{-3}, \ldots, 10^{4}\}$$

Read it

A coarse logarithmic sweep finds the scale before local refinement.

Boundary

A dense linear grid wastes trials at uninformative large values.

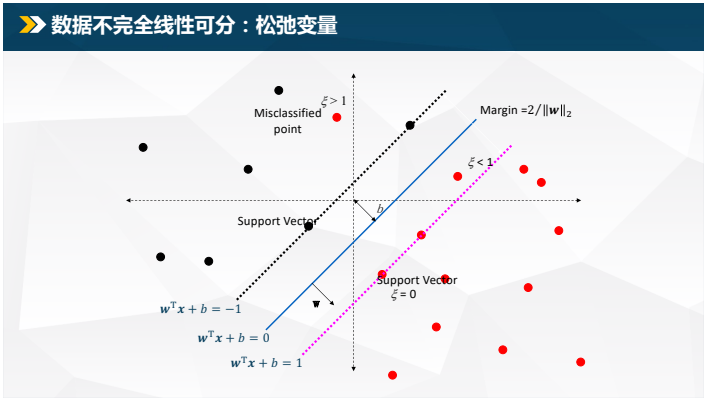
Support-vector count is a diagnostic

Lens	Evidence
Principle	Record the fraction of training cases with nonzero dual coefficients.
Case	A very high fraction may signal overlap, noise, or an overly local kernel.
Risk	Support count alone is not a selection metric.



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Soft margins tolerate some violations



Why introduce the dual problem

Start from evidence

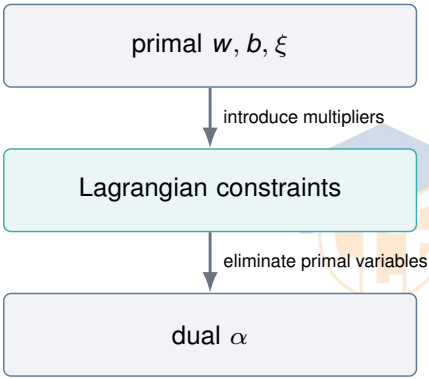
Kernel substitution becomes possible after $x_i^T x_j$ appears.

Model statement

The dual exposes sample coefficients and inner products.

Failure mode

Duality is not only a faster solver trick.



The dual sees samples only through inner products.

The hard-margin Lagrangian

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$$\mathcal{L} = \tfrac{1}{2}\|w\|^2 - \sum_i \alpha_i [y_i(w^T x_i + b) - 1]$$

Read it

Each inequality receives a nonnegative multiplier.

Boundary

The sign of a multiplier must match the constraint convention.

Stationarity reconstructs the weight vector

$$\frac{\partial \mathcal{L}}{\partial w} = 0 \Rightarrow w = \sum_i \alpha_i y_i x_i$$

Read it

The normal vector is a weighted sum of training samples.

Boundary

Cases with zero α_i disappear from the expansion.

The bias condition balances labels

$$\frac{\partial \mathcal{L}}{\partial b} = 0 \Rightarrow \sum_i \alpha_i y_i = 0$$

Read it

Positive and negative weighted contributions must balance.

Boundary

Dropping the intercept changes this equality.

The dual objective contains only inner products

$$\max_{\alpha} \sum_{i} \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j$$

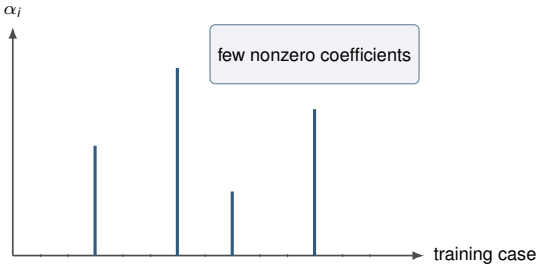
Read it

Subject to $\alpha_i \geq 0$ and $\sum_i \alpha_i y_i = 0$.

Boundary

The Gram matrix can be large when the sample count is large.

Dual coefficients are usually sparse



Start from evidence

Most cases far from the boundary receive coefficient zero.

Model statement

Only active constraints have positive multipliers.

Failure mode

Soft margins may produce many bounded coefficients at C.

Complementary slackness names the support vectors

$$\alpha_i[y_i f(x_i) - 1] = 0$$

Read it

If the margin constraint is strict, its multiplier must be zero.

Boundary

A point on the margin may still have a numerically tiny coefficient.

Soft-margin KKT adds an upper bound

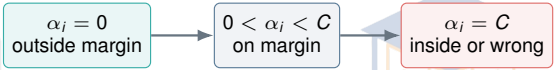
$$0 \le \alpha_i \le C$$

Read it

Coefficients at C correspond to strongly violating or boundary-limited cases.

Boundary

Numerical tolerances blur exact equality.



Complementary slackness converts algebra into sample roles.

Recover the intercept from free support vectors

$$b = y_i - \sum_j \alpha_j y_j K(x_j, x_i), \quad 0 < \alpha_i < C$$

Read it

Average the estimate over several free support vectors.

Boundary

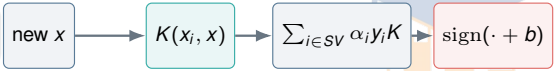
Using a bounded point at C can bias the estimate.

Prediction sums similarities to support vectors

$$f(x) = \sum_{i \in SV} \alpha_i y_i K(x_i, x) + b$$

Read it

A new case compares only with retained support vectors.



Boundary

Prediction time grows with the support-set size.

A hand calculation with two supports

$$f(x) = 0.8K(x_+, x) - 0.8K(x_-, x) + b$$

Read it

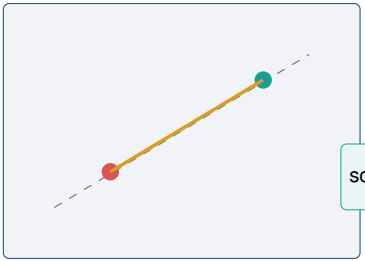
Compare one new point with one positive and one negative support.

Boundary

Similarity magnitudes depend on the chosen kernel and scaling.

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SMO updates two coefficients at a time



What changes

The equality constraint prevents changing only one α_i freely.

What can break

Poor pair selection can make convergence slow.

Try this

Choose a violating pair, solve its bounded quadratic, then update the bias.

The two-variable feasible set is a line segment

$$y_1\alpha_1 + y_2\alpha_2 = \zeta, \qquad 0 \le \alpha_1, \alpha_2 \le C$$

Read it

Intersect an equality line with the coefficient box.

Boundary

The endpoints differ for equal and opposite labels.

KKT violations guide working-set selection

Core idea

Select coefficients that most strongly violate optimality conditions.

Concrete case

Recheck error-cache values after an update.

Watch for

A stale cache produces incorrect priorities.

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Stopping tolerances are numerical decisions

- Convergence is declared when KKT residuals fall below a tolerance.
- Tighter tolerance can increase runtime with little validation benefit.
- A solver status message should be retained in experiment logs.

Primal or dual depends on data shape

Start from evidence

Kernel dual solvers are natural when pairwise similarity is essential.

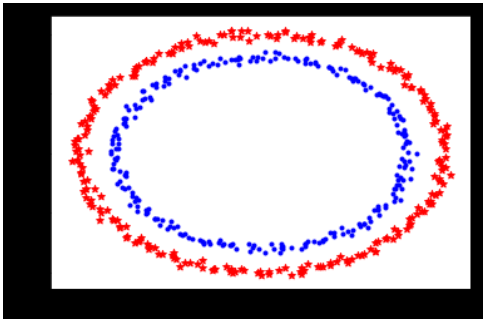
Model statement

Linear primal solvers are attractive when features are manageable and samples are numerous.

Failure mode

The word SVM does not imply one fixed optimizer.

A nonlinear pattern can become linearly separable



Core idea

Map inputs to a feature space where a hyperplane is useful.

Concrete case

Concentric rings are not separable by a line in the original plane.

Watch for

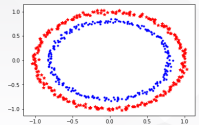
The mapped space may be very high or infinite dimensional.

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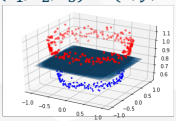
A feature map makes a curved pattern separable

核方法

- 前面我们用超平面（线性模型）来分开不同类的训练样本。
- 但在实际任务中，原始样本空间也许并不存在一个超平面能将训练样本分开。
- 例如：



$$(z_1, z_2, z_3) = (x, y, x^2 + y^2)$$



- 对这类问题，我们可以将原始空间映射到一个更高维特征空间，使得在这个特征空间数据线性可分。

The kernel trick replaces mapped inner products

$$K(x, z) = \phi(x)^T \phi(z)$$

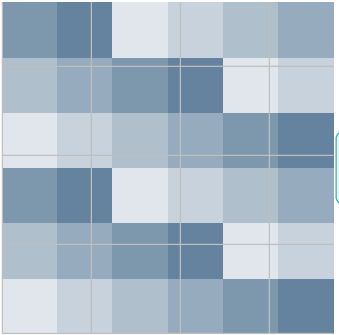
Read it

Compute similarity without materializing every mapped coordinate.

Boundary

Not every arbitrary similarity function is a valid kernel.

A Gram matrix stores all training similarities



symmetric

positive semidefinite

pairwise similarity

$$K_{ij} = K(x_i, x_j)$$

Read it

Rows and columns correspond to the same training cases.

Boundary

Memory grows quadratically with sample count.

Valid kernels are symmetric positive semidefinite

$$K = K^T, \qquad c^T K c \geq 0 \ \forall c$$

Read it

A positive semidefinite Gram matrix defines a consistent inner-product geometry.

Boundary

A hand-designed score may violate this property.

Kernel sums preserve validity

$$K = aK_1 + bK_2, \qquad a, b \geq 0$$

Read it

Combine text and metadata similarities with nonnegative weights.

Boundary

Negative weights can destroy positive semidefiniteness.

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The linear kernel is the baseline

$$K(x, z) = x^T z$$

Read it

It gives the ordinary linear SVM in dual form.

Boundary

A linear kernel does not eliminate the need for scaling.

A degree-two polynomial kernel includes interactions

For $x = (x_1, x_2)$ and $z = (z_1, z_2)$, take $K(x, z) = (x^T z)^2$.

$$(x_1 z_1 + x_2 z_2)^2 = x_1^2 z_1^2 + 2 x_1 x_2 z_1 z_2 + x_2^2 z_2^2.$$

An explicit feature map is

$$\phi(x) = (x_1^2, \sqrt{2}\, x_1 x_2, x_2^2), \qquad K(x, z) = \phi(x)^T \phi(z).$$

For $x = (1, 2)$ and $z = (3, 1)$:

$$K(x, z) = (3 + 2)^2 = 25 = 1 \cdot 9 + (2\sqrt{2})(3\sqrt{2}) + 4 \cdot 1.$$

Product $x_1 x_2$ is an interaction. Select the degree using validation data.

An explicit degree-two expansion

$$(x^T z)^2 = \langle \phi(x), \phi(z) \rangle$$

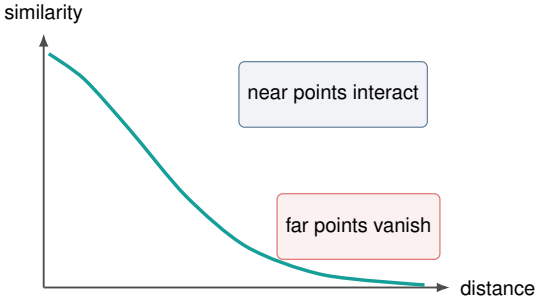
Read it

For two inputs, the map contains squared terms and a scaled cross term.

Boundary

The square-root scaling on cross terms matters for exact equality.

The RBF kernel measures local similarity



$$K(x, z) = \exp(-\gamma \|x - z\|_2^2)$$

Read it

Nearby points have similarity near one; far points approach zero.

Boundary

Distance is meaningless if feature scales are incompatible.

Gamma is inverse neighborhood width

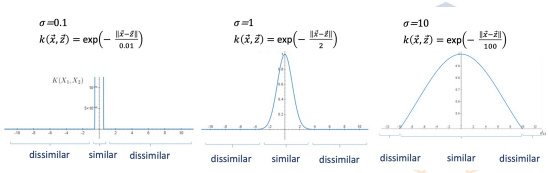
$$\gamma = \frac{1}{2\sigma^2}$$

Read it

Larger γ means a narrower influence region.

Boundary

Parameter names differ across libraries and texts.



Small gamma produces smooth boundaries

Core idea

Many distant points contribute appreciable similarity.

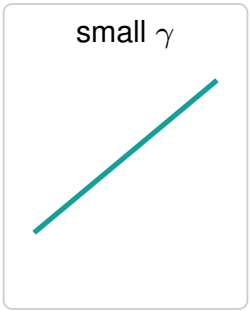
Concrete case

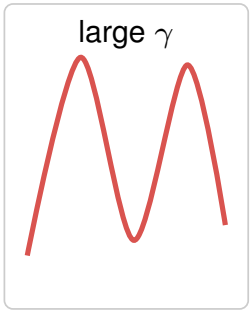
Use it when class structure changes gradually in scaled space.

Watch for

Too small a value can underfit local structure.

Large gamma produces local islands





- Only very near support vectors influence a prediction.
- Training accuracy can rise while validation stability falls.
- A fragmented boundary may memorize isolated noise.

Kernel width controls the locality of the boundary.

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C and gamma interact

Start from evidence

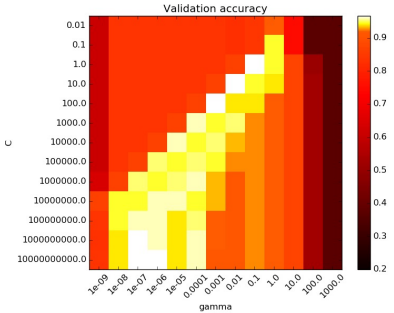
A ridge of good validation scores may run diagonally across the grid.

Model statement

High C and high γ can jointly create a highly flexible classifier.

Failure mode

Tuning one parameter while fixing the other can miss the useful region.



Read a C-gamma heatmap

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What changes

Each cell is a cross-validated score for one parameter pair.

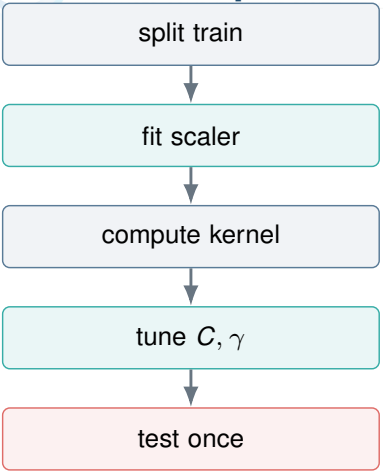
What can break

The brightest cell may be a noisy winner.

Try this

Prefer a stable plateau when scores are practically tied.

Nested validation protects the final estimate



Every transformation is learned inside the training fold.

Core idea

Tune kernel settings inside the inner loop and assess the whole procedure outside.

Concrete case

Every outer fold repeats scaling and parameter search.

Watch for

Reusing outer-test feedback leaks model-selection information.

A movie similarity story

Core idea

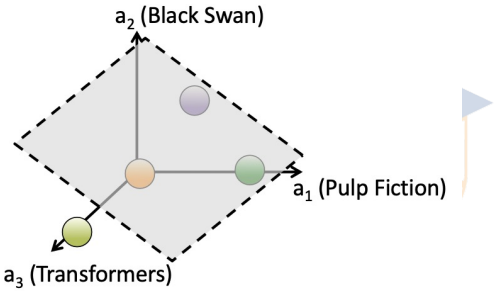
Represent each film with budget, rating, and genre-related features.

Concrete case

A Palme d'Or winner may be surrounded by films that are locally similar.

Watch for

Raw feature distances mix units and semantics.



RBF similarity decreases with distance

- $k(x, z) = \exp(-\gamma \|x - z\|^2)$, with $\gamma > 0$.
- At distance 0, the similarity is 1.
- For $\gamma = 1$, distances 1 and 2 give $e^{-1} \approx 0.368$ and $e^{-4} \approx 0.018$.
- Larger γ gives a narrower neighborhood; feature scales affect the distances.

Kernel centering matters for some methods

$$K_c = HKH, \qquad H = I - \frac{1}{N}\mathbf{1}\mathbf{1}^T$$

Read it

Center feature-space coordinates without constructing them explicitly.

Boundary

Fit centering statistics only on training data when used in a pipeline.

Kernel normalization controls self-similarity

$$\widetilde{K}(x,z) = \frac{K(x,z)}{\sqrt{K(x,x)K(z,z)}}$$

Read it

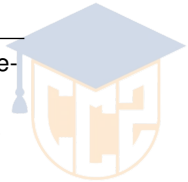
Useful when vector magnitudes should not dominate direction-like similarity.

Boundary

Normalization changes the model and must be validated.

Categorical inputs need an explicit design

Lens	Evidence
Principle	One-hot encoding plus a linear or RBF kernel is one baseline.
Case	A custom equality kernel can compare category matches.
Risk	Integer category codes create fake distances.



Text often favors sparse linear SVMs

Core idea

High-dimensional sparse counts can already be linearly separable enough.

Concrete case

TF-IDF with a linear SVM is a strong document baseline.

Watch for

Dense RBF Gram matrices may be unnecessary and expensive.

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Image descriptors may benefit from nonlinear similarity

- Kernel choice should reflect the invariances encoded by the representation.
- Compare linear and RBF kernels after a fixed feature extractor.
- A better kernel cannot repair a representation that discards the signal.

One-vs-rest builds one classifier per class

Start from evidence

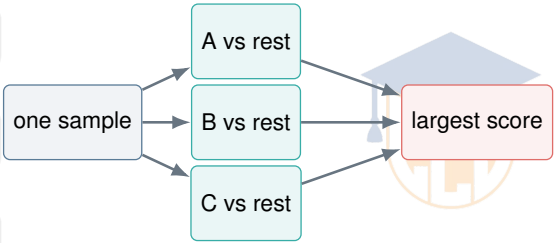
Choose the class with the largest comparable decision score.

Model statement

Class k is positive and all remaining classes are negative.

Failure mode

Independent binary scores are not calibrated probabilities by default.



One-vs-one builds pairwise classifiers

$$\frac{K(K-1)}{2} \text{ binary models}$$

Read it

Each pair votes or contributes a coupled score.

Boundary

The number of models grows quadratically with class count.

The Otto case uses multiclass log loss

Lens	Evidence
Principle	Probabilistic output quality matters beyond the winning class.
Case	Nine product categories require reliable ranking across all labels.
Risk	A raw SVM margin is not a probability.



Probability calibration is a second model

Core idea

Platt scaling or isotonic regression maps scores to probabilities.

Concrete case

Fit calibration on held-out or cross-validated predictions.

Watch for

Calibrating on the same fitted scores is optimistic.

Kernel approximation can scale RBF ideas

- Random Fourier features or Nyström features create explicit finite embeddings.
- Train a linear classifier on the approximate map.
- Approximation rank trades speed against fidelity.

Kernel matrix memory grows quadratically

$$N = 50{,}000 \Rightarrow N^2 = 2.5 \times 10^9 \text{ entries}$$

Read it

At eight bytes per entry, the raw matrix is about twenty gigabytes.

Boundary

Solver overhead adds more memory beyond the matrix.

A short kernel experiment

```
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
X, y = make_classification(n_samples=320, n_features=4, random_state=7)
X_train, X_valid, y_train, y_valid = train_test_split(X, y, random_state=7)
model = make_pipeline(StandardScaler(), SVC(kernel='rbf', C=10, gamma=0.1))
model.fit(X_train, y_train)
print(model.score(X_valid, y_valid))
```

Expected observation

Compare linear and RBF validation scores under the same split.

Check

Do not scale the validation data separately.

Inspect support vectors after fitting

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```
svc = model.named_steps['svc']
print(svc.n_support_)
print(svc.support_.shape)
```



Expected observation

Report support counts by class and total support fraction.

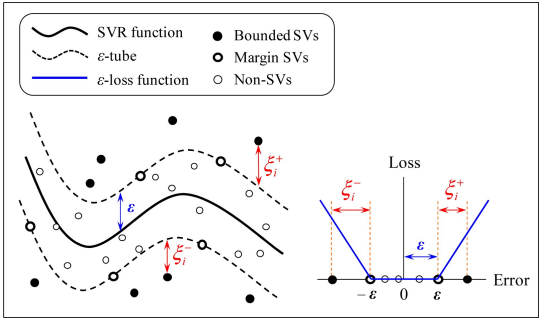
Check

Indices refer to the transformed training order.

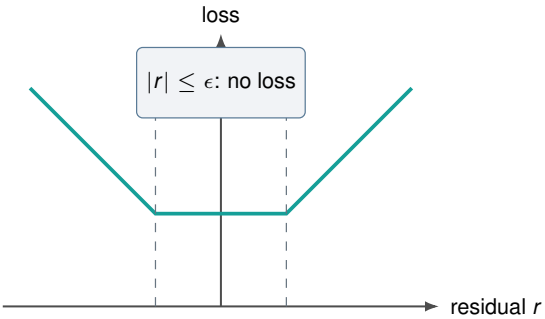
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Regression replaces a margin corridor with a tube

- Fit a function so that many targets lie within $\pm\epsilon$.
- Small deviations inside the tube are ignored.
- The tube width is a modeling tolerance, not a confidence interval.



Epsilon-insensitive loss has a flat center



$$L_\epsilon(r) = \max(0, |r| - \epsilon)$$

Read it

A residual of 0.3 pays no loss when $\epsilon = 0.5$.

Boundary

Residuals outside the tube pay only the excess beyond ϵ.

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Compare regression loss families

What changes

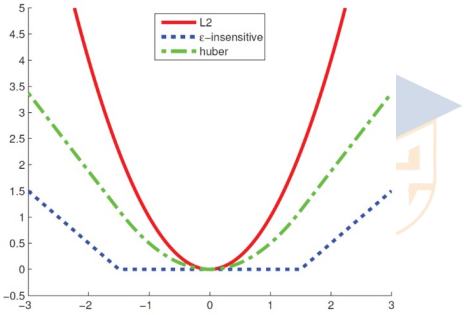
Squared loss grows quadratically; Huber changes regime; epsilon loss ignores a central band.

What can break

Losses with similar curves near zero can behave very differently on large residuals.

Try this

Choose the loss from the error cost, not from habit.



Linear SVR primal objective

$$\min_{w,b,\xi,\xi^*} \frac{1}{2}\|w\|^2 + C\sum_i (\xi_i + \xi_i^*)$$

Read it

Use separate slacks above and below the tube.

Boundary

The two sides need not have different costs unless the task requires asymmetry.

Two inequalities define each regression tube

$$y_i - f(x_i) \le \epsilon + \xi_i, \quad f(x_i) - y_i \le \epsilon + \xi_i^*$$

Read it

At most one side is active for an ordinary residual.

Boundary

Incorrect sign conventions swap upper and lower slacks.

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Support vectors sit on or outside the tube

- Points strictly inside the tube have zero effective dual coefficient.
- Boundary and outside points determine the fitted function.
- A target can be a support vector without being an extreme outlier.



Dual SVR uses coefficient differences

$$f(x) = \sum_i (\alpha_i - \alpha_i^*) K(x_i, x) + b$$

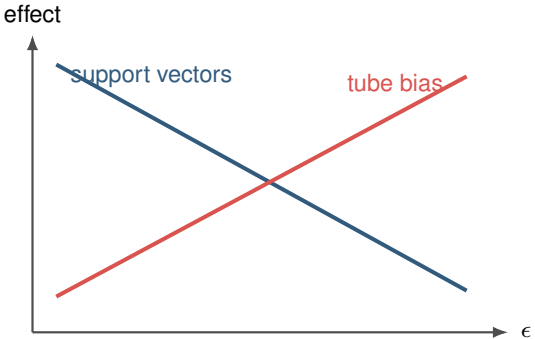
Read it

Upper and lower violations contribute with opposite signs.

Boundary

Both coefficient families are bounded by C.

Epsilon controls sparsity and bias



What changes

Larger ϵ widens the no-penalty tube and usually reduces support count.

What can break

A wide tube can hide practically important errors.

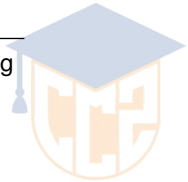
Try this

Choose epsilon in units of the target after any transformation.

C controls outside-tube error

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Lens	Evidence
Principle	Large C bends the function more strongly toward violating targets.
Case	Compare validation MAE and support fraction across C.
Risk	High C with high gamma can overfit individual targets.



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Target scaling changes epsilon meaning

- Scaling y requires scaling or retuning ϵ and interpreting metrics back in original units.
- An epsilon of 0.1 means different things for standardized and raw demand.
- Do not compare raw epsilon across target transformations.

SVR is not safe for extrapolation

Start from evidence

Future temperatures beyond the training range have weak similarity to every support vector.

Model statement

RBF predictions far from all training cases tend toward the intercept-like baseline.

Failure mode

A smooth validation curve inside range does not prove extrapolation reliability.

Time-aware validation for bike demand

What changes

Train on earlier dates and validate on later dates.

What can break

Random shuffling leaks neighboring time patterns.

Try this

Weather, holidays, and seasonality create temporal structure.

Bike-demand tuning is a surface

Lens	Evidence
Principle	Evaluate C, γ, and ϵ jointly with a deployment metric.
Case	Record runtime and support count beside MAE.
Risk	Optimizing three parameters on one validation set increases selection bias.

Fit SVR in a leak-safe pipeline

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```
from sklearn.datasets import make_regression
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVR
X, y = make_regression(n_samples=320, n_features=4, noise=12, random_state=7)
X_train, X_valid, y_train, y_valid = train_test_split(X, y, random_state=7)
model = make_pipeline(StandardScaler(), SVR(C=10, epsilon=0.1, gamma='scale'))
model.fit(X_train, y_train)
print(model.score(X_valid, y_valid))
```

Expected observation

Keep scaler fitting inside every training fold.

Check

The default score is R^2, not MAE.

Report MAE and residual structure

$$\mathrm{MAE} = \frac{1}{N}\sum_i |y_i - \hat{y}_i|$$

Read it

Add a residual-versus-prediction plot and subgroup summaries.

Boundary

One average can hide seasonal bias.

Sample size changes the visible margin

What changes

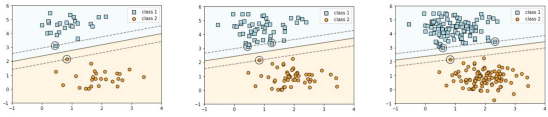
With more observations, the estimated class frontier becomes better constrained.

What can break

More duplicated or biased data do not improve representativeness.

Try this

Compare the same data-generating process at $N = 50$, 100, and 200.



A classifier comparison must share evidence

Lens	Evidence
Principle	Use identical splits, preprocessing, metrics, and search budgets.
Case	Compare linear SVM, RBF SVM, and logistic regression on one pipeline.
Risk	Giving one model a larger tuning budget biases the contest.

Check data and model settings before changing the kernel

1. Check labels and duplicates.
2. Inspect feature units and preprocessing.
3. Match the split to groups or time.
4. Only then compare kernels and parameters.

Use case

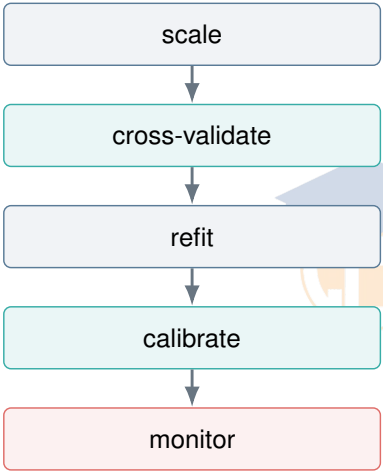
An RBF kernel cannot repair target leakage.

Stop if

Complexity can make a flawed pipeline look more convincing.

A deployment card for SVM

Item	Record
Input	features, units, missing values
Model	kernel, C, γ, class weights
Evidence	split, metric, uncertainty
Runtime	support count and latency
Monitoring	range, drift, error slices



A strong margin is only one component of a reliable system.

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Open-book exam case

What changes

A high-C, high-γ SVM reaches perfect training accuracy but unstable grouped validation.

What can break

A strong answer connects every action to observed evidence.

Try this

Diagnose likely overflexibility and possible group leakage; propose two controlled experiments.

SVM self-check

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1. Derive why maximizing margin minimizes $\|w\|^2$ after scaling.
2. Interpret the three slack regimes.
3. Explain why kernels appear naturally in the dual.
4. Predict how C, γ, and ϵ affect complexity.
5. Name one leak-safe validation design.

SVM in one sentence

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Core idea

SVM learns a regularized boundary or tube whose prediction depends on a subset of critical training cases.

Concrete case

Linear, kernel, classification, and regression versions share this structure.

Watch for

A sophisticated margin cannot compensate for invalid data evidence.